MIL-E-1/738C
14 August 1968
SUPERSEDING
MIL-E-1/738B
24 September 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 2C42

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Planar triode

See figure 1

Mounting position: Any

Weight: 3 ounces nominal

ABSOLUTE RATINGS:

Anode pulsed oscillator:

Parameter:	© Fl	Ef	eb	Eb	ik	Pp	tp	Ck	Rk	tk	T(seal)	© Ehk
Unit:	MHz	V	v	V	a	W	μs	μF	Ohms	sec	°C	Vdc
Maximum:	1,300	6.6	2,000	---	1.5	25	9	---	---	---	200	±90
Maximum:	1,300	6.6	3,000	---	2.5	12	3	---	---	---	200	---
Minimum:	---	6.0	---	---	---	---	---	---	---	60	---	---
TEST CONDITIONS:	---	6.3	---	250	---	---	---	1,000	100	300	---	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) t = 168 hours

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
Appendix D, 30(a), 40	Visual and mechanical inspection criteria		---	---	---	---	---	---
1236	Power oscillation	eb = 3,000 v; Rg = 50 ohms; F = 1,050 ± 50 MHz; tp = 1.0 µs; prr = 1,000 (see note 4) (Dwg 146-JAN)	0.65	II	Po	1.2	---	W
1231	Pulse emission	prr = 500 ± 10%; tp = 2.0 µs; (eb = ec)/is = 4.0 a	0.65	II	etd	---	190	v
1256	Electrode current (anode)		0.65	II	Ib	12	31	mAdc
1306	Transconductance		0.65	II	Sm	6,200	10,000	µmhos
1266	Total grid current		0.65	II	-Ic	---	0.5	µAdc
	Quality conformance inspection, part 2							
Ⓒ Appendix D, 30(b)	Dimensions	See figure 1	6.5	S3	---	---	---	---
1231	Emission	(Eb - Ec)/Is = 40 mAdc	6.5	S3	Etd	---	8.0	Vdc
1211	Insulation of electrodes	E(g to all) = -100 Vdc E(sh to all) = -500 Vdc	6.5	S3	R	100	---	Meg
1301	Heater current		6.5	S3	If	840	960	mAdc
Ⓒ 1261	Electrode voltage (cutoff) (grid)	Ec/Ib = 10 µAdc	6.5	S3	-Eco	---	24	Vdc
1336	Heater-cathode leakage	Ehk = ± 100 Vdc	6.5	S3	Ihk	---	50	µAdc
1331	Direct-interelectrode capacitance	No voltages (DESC Dwg 65025)	6.5	S3	Cgp Cin Cout Cksh	1.60 2.45 --- 50	1.90 3.25 0.040 200	pF pF pF pF
1316	Amplification factor		6.5	S3	Mu	30	70	---
	Quality conformance inspection, part 3							
1031	High-frequency vibration	F = 50 Hz; 10 G; Ebb = 300 Vdc; Rp = 10,000 ohms; Ec/Ib = 10 mAdc (see notes 2 and 4)	---	---	Ep	---	100	mVac
---	Torque	No voltages (see notes 2 and 3)	---	---	---	---	---	---
---	Torque-test end point	Total grid current	---	---	-Ic	---	0.5	µAdc
---	Frequency drift	See notes 1, 2, and 5	---	---	ΔF	---	4.5	MHz

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued							
Ⓒ ---	Life-test provisions	eb = 2,000 v; Rg = 50 ohms; tp = 3 ± 0.3 μs; prr = 2,000 ± 5%; F = 1,050 ± 50 MHz; group A (see note 1) (Dwg 146-JAN)	---	---	t	500	---	hrs
---	Life-test end point	Power oscillation	---	---	Po	0.8	---	W

NOTES:

1. The voltage pulse applied to the tube under test shall have the following characteristics measured with the pulse supplying power to a noninductive resistor of 1,000 ohms in place of the tube: tp = 1 ± 0.1 μs, measured at 80 percent of mean amplitude of pulse; tr to be less than 0.2 μs; tf to be less than 0.4 μs; variations in amplitude over 80 percent if the top portion shall be not more than ± 5 percent of the mean amplitude.

2. Test 10 tubes selected at random from the first production lot of each calendar year, and approximately every 90 days of production during the year. If more than one tube fails to pass the specified end point, the failed test shall become a part of quality conformance inspection, part 2, on all lots in process. After three consecutive successful submissions, the test shall revert back to the 10-tube quarterly test.

Ⓒ 3. Apply torque of 20 lb-in between anode and pins without shock.

4. Vibrate in direction perpendicular to plane of grid.

5. Measure frequency under condition "A" after the tube has become stable, and again under condition "B" when the tube has become stable. The difference in these two frequencies is defined as the frequency drift.

	Condition "A"	Condition "B"
eb	2,200 volts	1,800 volts
Ef	6.9 ± 0.1 V	5.7 ± 0.1 V
prr	5,000 ± 5%	500 ± 5%
TA	30° ±5°C	80° ±5°C

Use standard frequency drift test, in accordance with CRG Model No. TP34AX, or equivalent.

The margins of this tube specification sheet are marked with a revision letter Ⓒ to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

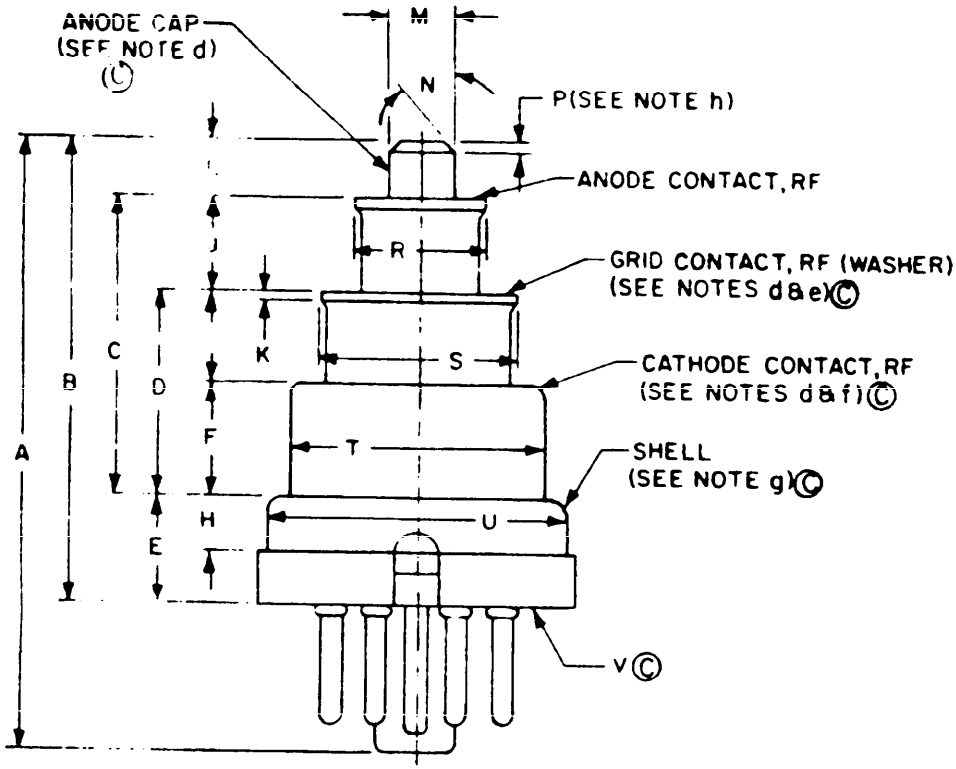
Preparing activity:
Air Force - 11

(Project 5960-2318-2)

Review activities:
Army - EL
Navy -
Air Force - 11, 17, 85
DSA - ES

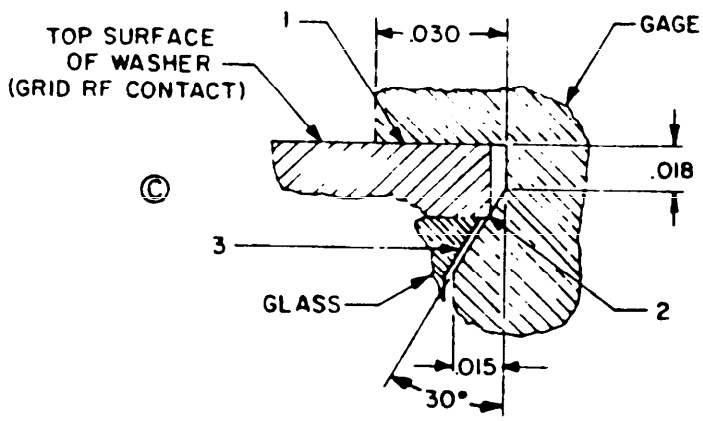
User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 19

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PIN CONNECTIONS

Pin No.	Element
1	int. con.
2	h
3	k
5	k
7	h
8	k
cap	a ©



With top surface of washer (grid rf contact) in contact with surface (1) of gage and with lower edge of washer (grid rf contact) in contact with surface (2) of gage, external glass surface (3) of tube shall not be in contact with gage at any point. Equivalent gaging procedure may be used.

GAGING PROCEDURE, GLASS SURFACE

FIGURE 1. Outline drawing of electron tube type 2C42.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
© B		2.078 (52.78)
C	1.220 (30.99)	1.260 (32.00)
D	.850 (21.59)	.880 (22.35)
F	.475 (12.07)	.505 (12.83)
G	.360 (9.14)	.390 (9.91)
L	.347 (8.81)	.363 (9.22)
M	.371 (9.42)	.379 (9.63)
S	.808 (20.52)	.816 (20.73)
T	1.023 (25.98)	1.039 (26.39)
(See note c)		
© A		2.688 (68.27)
E	.395 (10.03)	.455 (11.56)
H	.180 (4.57)	.210 (5.33)
J	.360 (9.14)	.390 (9.91)
K	.030 (.76)	.035 (.89)
P	.016 (.41)	.047 (1.19)
R	.557 (14.15)	.567 (14.40)
© U	1.216 (30.89)	1.236 (31.39)
N	45° (see note b)	
© V	Base: B6-108)	

NOTES:
a. Dimensions are in inches, unless otherwise specified.
b. These dimensions are for information only, and are not required for inspection purposes.
c. These dimensions shall be inspected on 10 tubes a month when in continuous production. Failure of more than one tube to meet tolerances for any dimension shall cause that dimension to become for all lots in process, part of quality conformance inspection, part 2.
d. Eccentricity of contact surfaces shall be as follows: Quality conformance inspection, part 2, shall apply.

Contact surface	TIR, maximum	Reference surface
Grid	.016	Anode cap
Cathode, rf	.016	Grid

e. External glass surface of tube shall conform with the detailed requirements of figure 1. Quality conformance inspection, part 2, shall apply.
f. Cathode rf contact surface shall be free from welding flash material, quality conformace inspection, part 2, shall apply.
g. Limits of dimension U do not apply at points where shell is crimped to base.
h. Radius with limits as specified may be used instead of chamfer.
j. External metal surfaces, except base pins, shall be silver plated 100 msi, minimum. Quality conformance inspection, part 2, shall apply.

FIGURE 1. Outline drawing of electron tube type 2C42 - Continued.

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